Dräger Perseus A500 Anaesthesia Workstation

Outstanding ventilator technology meets the latest approaches to ergonomics and system integration in one innovative anaesthesia machine, developed together with experts from all over the world to streamline your anaesthesia workflow.

Dräger

Technology for Life

Benefits

Advanced, economical ventilation technology

The Perseus A500 lets you achieve high-quality ventilation for individualised ventilation strategies, always facilitating and supporting the patient's spontaneous breathing. Lung recruitment functions provide manoeuvres to automate various operating sequences. You can adjust and individually control these at any time. Because of the optimised breathing system, changes in the gas concentration reach the patient more readily, especially during low- and minimal-flow anaesthesia.

Flexible workflow support

The workflow-support features of the Perseus A500 are designed to streamline and simplify routine tasks. The time-controlled, fully automatic system test also includes innovative technologies like an O_2 real-gas test and automatic recognition of any incorrect connected breathing hoses. Other advantages include seamless monitoring transfer from bedside to the OR with a single monitor. The clinical situation oriented emergency start-up ventilation functions, incl. APRV, work even when the machine is switched off. The data analysis function enables a straightforward export of data by using a USB flash drive. In addition, the Perseus A500 uses RFID technology and can inform you when the individual Infinity ID accessories need to be replaced. Because of the many mounting possibilities for the holder arm and monitors, the Perseus A500 can be configured to meet individual needs. The variety of many remote service options, facilitates you to implement individualised remote maintenance concepts.

Prediction of inspiratory and expiratory concentrations of volatile anaesthetics

The Perseus A500 is compatible with Vapor 2000 and Vapor 3000 with auto-exclusion connection system. Combined with the VaporView option (and Vapor 3000/D-Vapor 3000), Perseus A500 provides you sophisticated oxygen and anaesthetic agent level and xMAC prediction technology. This allows you to administer low- and minimal-flow anaesthesia more intuitively.

Enhanced workplace ergonomics

The Perseus A500 contains many features that significantly improve workplace ergonomics. It has a spacious, well-lit work surface with plenty of storage space for consumable material. In addition, the optimally positioned central brake, suction unit and anaesthetic gas scavenging system allow you to use the Perseus A500 easily and intuitively. The integrated breathing system can be opened without tools and prepared for reprocessing.

Supportive design

Despite the Perseus A500's completely new design, the user interface utilizes the familiar Dräger operating system featuring the same rotary knob you know from other Dräger ventilators and anaesthesia devices. Therefore, using the Perseus A500 is as straightforward as with any other Dräger medical device. The sleek, modern design makes the machine into an extremely flexible workspace that helps you simplify your workflow while maintaining the highest quality of therapy.

Outstanding design

The Perseus A500 has been awarded two major design prizes: the 'iF Product Design Award 2013' in the category of medicine / health + care, and the "Red Dot Design Award 2013: Best of the Best" in the category of life sciences and medicine. Both awards count among the most important international design competitions and not only rate design quality, but also aspects such as safety, ergonomics, functionality, degree of innovation and, not least, environmental compatibility.

System Components


D-7420-2011

Dräger Vapor 3000 and D-Vapor 3000

As all other Dräger vaporisers, the Dräger Vapor 3000 series and D-Vapor 3000 deliver same remarkable performance when it comes to precise agent delivery, safety, robustness, quality, and durability. In combination with the Perseus A500 anaesthesia workstation, they additionally can contribute to improved workflow efficiency, clinical outcome, and decision-making support.


D-30739-2017

Infinity Acute Care System

Transform your clinical workflow with Infinity Acute Care System. Its multiparameter monitor integrates with its networked medical-grade workstation, giving you real-time vital signs, access to clinical hospital systems and data management applications for a comprehensive range of patient information and powerful analysis tools at the point-of-care.


D-45302-2021

Dräger SmartPilot View

The software estimates and visualises complex interactions of anaesthetic drugs based on pharmacodynamic modelling for both the current and predicted course of anaesthesia. SmartPilot View supports informed decision-making that facilitates more precision and individualisation in anaesthetic drugs titration.

Accessories

D-14586-2009

Infinity ID-accessories

Each and every Infinity ID-accessory has been designed to offer additional functionality, which can help you simplify routine tasks, streamline workflow and increase safety levels.

D-14348-2017

WaterLock 2

Protects your patient, protects your gas measurement systems. Designed to give you reliable gas measurements, the Dräger WaterLock 2 helps you to effectively filter humid and contaminated exhaled air thanks to our advanced membrane technology. Keeping your patients and investments safe from water, bacteria, and potential viruses.

D-6414-2018

Drägersorb 800+ – Soda Lime

Click and connect with 100% reliability. As one of the leading manufacturers of anaesthesia equipment, we believe in leading the way to producing high-quality soda lime that ensures your patients' and staff's safety to the highest degree. Drägersorb is more than just a formula, it is absorption efficiency — you can trust.

Related Products

D-9003-2016

Zeus Infinity Empowered

Lung-protective ventilation functionalities, cost-saving gas delivery technology, decision-support tools and comprehensive monitoring. Zeus IE integrates them all into one advanced anaesthesia workstation. It can, hence, improve clinical outcome, workflow efficiency, and cost savings throughout the anaesthesia process.

D-2308-2022

Dräger Atlan A350/A350 XL

The new platform offers flexibility for most spatial conditions. The high precision piston ventilator supports lung protective ventilation measures and a comprehensive set of parameters assist decision-making support. The Atlan A350/XL can be networked to communicate securely with other networked devices to share data and information that can help to increase efficiency and reduce errors in anaesthesia.

Technical Data

Technical data	
Weight	Approx. 160 kg (basic setup)
Dimensions	(H x W x D) 148 cm x 115 cm x 79 cm (58.3 in x 45.2 in x 31.1 in)
Power consumption	70 W, typical, max. 2.2 kW with auxiliary power outlets in use
Mains power supply	100 – 127 V ~ 50/60 Hz or 220 – 240 V ~ 50/60 Hz
Maximum power consumption	12 A
Integrated battery backup time	Min. 30, typically 150 minutes (with new and fully charged batteries)
Data interfaces	2 x RS 232 (MEDIBUS Protocol), 1 x USB, 1 x LAN
Integrated power sockets	4 x country-specific (with isolation transformer) or 4 x IEC
Storage surface and drawers	1 drawer with lock and writing surface (optional)
	2 additional drawers (optional), including one with lock
Working surface	Approx. 85 x 35-50 cm (suitable for DIN A3 paper format)

Application and ambient conditions	
Temperature	10 to 40 °C (50 to 104 °F)
Air pressure	620 to 1,060 hPa (9.0 to 15.3 psi) equivalent to 4,000 metre elevation

Fresh-gas delivery – electronic mixer	
Fresh-gas flow	OFF; 0.2 to 15 L/min
Dosable O_2 concentration	21 to 100 % in air; 25 to 100 % (in N_2O)
O_2 flush	25 to 75 L/min at 2.7 to 6.9 bar gas supply pressure
O_2 flow for auxiliary and additional oxygen O_2 emergency delivery	OFF; 2 to 10 L/min O_2 safety flow is also fed through the Vapor when the device is switched off

Breathing system (heated)	
Volume	approx. 2.2 L (incl. CO_2 absorber)
Absorber volume	approx. 1.2 - 1.5 L
Reprocessing	cleaning, disinfection and sterilisation; replaceable without additional tools
Number of individual components during reprocessing	10
Heated breathing system, exchangeable without additional tools	

Technical Data

Ventilator

TurboVent2 Ventilator (electrically driven and electronically controlled turbo ventilator), fresh-gas decouled, ventilation also possible without any gas supply (driving gas consumption 0 L/min), autoclavable	
Standard ventilation modes	Manual/Spontaneous (MAN/SPON)
	Pressure-controlled: time-cycled (PC-CMV), synchronised (PC-BIPAP)
	Volume-controlled: time-cycled (VC-CMV), synchronised (VS-SIMV, time-cycled AutoFlow (VC-CMV/AF, synchronised AutoFlow (VC-SIMV/AF)
Optional ventilation modes	Pressure support ventilation (CPAP/PS), selectable pressure support for volume-controlled ventilation (VC-SIMV/PS), pressure-controlled ventilation (PC-BIPAP/PS), and AutoFlow (VC-SIMV/AF/PS), selectable CPAP for Manual/Spontaneous
	Airway Pressure Release Ventilation (PC-APRV)
	External fresh-gas outlet
	Insp./Exp. hold, lung-recruitment manoeuvre (One-step or Multi-step)
Patient demographics	Neonates, children, adults
Tidal volume	20 to 2000 mL in volume-controlled ventilation
	3 to 2500 mL in pressure-controlled ventilation
Inspiratory pressure (Pinsp)	PEEP +3 to 80 hPa (cmH_2O) (3 to 80 hPa (cmH_2O) when PEEP = Off)
Pressure limitation (Pmax)	PEEP +7 to 80 hPa (cmH_2O) (7 to 80 hPa (cmH_2O) when PEEP = Off)
Pressure support above PEEP (ΔPsupp)	Off, 1 to (80 - PEEP) hPa (cmH_2O)
Respiratory rate (RR)	3 to 100 /min
Inspiratory time (Ti)	0.2 to 10 s
Inspiratory flow	0 to 180 L/min
PEEP/CPAP	Off, 2 to 35 hPa (cmH_2O)
I:E	1:50 to 50:1
Triger sensitivity	0.3 to 15 L/min

Technical Data

Measuring systems, displays and other functionalities
15.3" (38.9 cm) touchscreen, configurable screen contens, smart alarm management with extensive support system
Minute volume (MV) and tidal volume (VT and ΔVT); Respiratory rate (RR); Peak inspiratory pressure (PIP); Plateau pressure (Pplat); Mean airway pressure (Pmean); Positive end-expiratory pressure (PEEP); Compliance; Resistance; MV x CO_2, O_2 uptake
Inspiratory and expratory gas concentration of O_2, N_2O, CO_2 and anaesthetic agents (automatic identification of halothane, entflurane, isoflurane, sevoflurane, desflurane); Age-corrected xMAC display; Vaporiser setting (optional); Prediction of anaesthetic gas concentration (optional); Prediction of inspiratory O_2 concentration (optional)
Simultaneous display of 3 or 4 real-time waveforms for: Concentration of CO_2, O_2 and anaesthetic agents, airway pressure, inspiratory and expiratory flow
Bar diagram display of volume and tidal volume; Virtual flow tubes for O_2, AIR, N_2O
Simultaneous display of 2 loops; Volume-pressure and flow-volume, reference loop
Display of grafical or tabular trends or mini-trends simultaneously with real-time waveforms and volume-pressure loop
Econometer for displaying fresh-gas efficiency (optionally including temporal trend or in the form of low-flow assistant)
Determination of consumption and uptake (determination of uptake only for anaesthetics) fresh-gas and anaesthetics per case and since last zeroing
AutoSet for alarm limits
Status display for display of airway pressure, supply status of battery and gases (CGS + cylinders)
Dosing of O_2 and anaesthetic agents during MAN/SPON ventilation possible, even when device is switched off
Programmable, time-based fully automatic start-up and system test of device and software including calibration of all sensors; normally no user action necessary after start of test
Integrated, dimmable illumination of working and documentation surfaces, illuminated vaporisers (optional)
Central brake, smooth running castors with cable detectors
Data storage on USB flash drive (alarm history, system test results, screen shots, trends and machine configurations; optionally: log files)
Free, six-week tral version of all available software options, activated individually by Dräger representative. Option expires automatically after end of trial period.

Corporate Headquarters
Drägerwerk AG & Co. KGaA
Moislinger Allee 53–55
23558 Lübeck, Germany
www.draeger.com

Region Europe
Drägerwerk AG & Co. KGaA
Moislinger Allee 53–55
23558 Lübeck, Germany
+49 451 882 0
+49 451 882 2080
info@draeger.com

Region Asia Pacific
Draeger Singapore Pte. Ltd.
61 Science Park Road
The Galen #04-01
Singapore 117525
+65 6872 9288
+65 6259 0398

Region Middle East, Africa
Drägerwerk AG & Co. KGaA
Branch Office
Dubai, United Arab Emirates
+971 4 4294 600
+971 4 4294 699
contactuae@draeger.com

Region Central and South America
Dräger Indústria e Comércio Ltda.
Al. Pucuruí 51 - Tamboré
06460-100 - Barueri - São Paulo
+55 (11) 4689-4900
relacionamento@draeger.com

Manufacturer
Drägerwerk AG & Co. KGaA
Moislinger Allee 53-55
23542 Lübeck, Germany

Locate your Regional Sales Representative at:
www.draeger.com/contact